

iClusterBayes Summary Report

Automated Report

Hyperparameters

```
cat('sdev = 0.05, beta.var.scale = 0.3\n')
```

```
## sdev = 0.05, beta.var.scale = 0.3
```

Key Summary Statistics

```
cat(paste0('```\n', '
--- Summary for K = 1 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.2217  0.2417  0.2483  0.2496  0.2572  0.2856

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.0000  0.5761  0.7175  0.6618  0.8056  0.9761
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.0000  0.5622  0.6972  0.6435  0.7839  0.9422
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.01556 0.04556 0.07111 0.23779 0.26667 1.00000
```

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Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.02278 0.04556 0.06778 0.20864 0.20625 1.00000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.01722 0.05278 0.13556 0.33933 0.63833 1.00000

--- Summary for K = 2 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.03944 0.05444 0.05833 0.05865 0.06278 0.08167

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.3979 0.5628 0.5163 0.6806 0.9517
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.3889 0.5483 0.4980 0.6567 0.9111
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01778 0.43722 0.50325 1.00000 1.00000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.007778 0.045556 0.329230 0.857083 1.000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.6083 1.0000 0.7680 1.0000 1.0000

--- Summary for K = 3 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.004444 0.012778 0.015556 0.015614 0.018333 0.030000

```

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beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.2817  0.4533  0.4189  0.5806  0.9178
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.2840  0.4528  0.4107  0.5689  0.8500
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 1.0000  1.0000  0.8456  1.0000  1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0005556 0.0097222 0.2908248 0.7755556 1.0000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.5583  1.0000  0.7613  1.0000  1.0000

--- Summary for K = 4 ---
Z.ar:
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.003889 0.011667 0.015556 0.017287 0.021667 0.050000

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.1289  0.3292  0.3133  0.4744  0.8950
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :

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Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1344	0.3344	0.3085	0.4700	0.8217

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.9489	1.0000	0.8198	1.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0005556	0.2987883	0.9131944	1.0000000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000	1.000	1.000	0.865	1.000	1.000

--- Summary for K = 5 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.001667	0.007778	0.010000	0.010944	0.013333	0.065000

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0750	0.2594	0.2602	0.4044	0.8633

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0800	0.2689	0.2584	0.4061	0.7694

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.4017	0.9417	0.6898	1.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
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    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000  0.0000  0.2785  0.7489  1.0000
Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 1.0000  1.0000  0.9043  1.0000  1.0000

--- Summary for K = 6 ---
Z.ar:
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.001667 0.005000 0.006667 0.007303 0.008889 0.029444

beta.ar:
Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0500  0.2139  0.2269  0.3572  0.8489
Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1
Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1
Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1
Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1

gamma.ar:
Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.05444 0.22167 0.22363 0.35778 0.72056
Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000  0.4206  0.4813  1.0000  1.0000
Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.575  1.000  1.000  1.000  1.000  1.000
Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000  0.0000  0.2445  0.4968  1.0000
Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.8367  1.0000  0.8093  1.0000  1.0000

--- Summary for K = 7 ---
Z.ar:
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.001667 0.005556 0.007222 0.007894 0.009444 0.031667

beta.ar:
Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

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0.00000 0.01389 0.19167 0.20592 0.33889 0.82556
Modality 2 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
Modality 3 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1

gamma.ar:
Modality 1 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.00000 0.01556 0.20333 0.20642 0.34778 0.69167
Modality 2 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000  0.0000  0.2017  0.4351  1.0000  1.0000
Modality 3 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.5033  1.0000  1.0000  0.9999  1.0000  1.0000
Modality 4 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000  0.0000  0.0000  0.2216  0.2803  1.0000
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000  0.7094  1.0000  0.7798  1.0000  1.0000

--- Summary for K = 8 ---
Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.001667 0.005556 0.006667 0.007356 0.008889 0.020556

beta.ar:
Modality 1 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.002778 0.107222 0.148832 0.253333 0.771667
Modality 2 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
Modality 3 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1

```

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.003333 0.117222 0.151073 0.268194 0.632778
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000  0.0000  0.3722  0.4600  1.0000  1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000  1.0000  1.0000  0.9964  1.0000  1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000  0.000  0.000  0.139  0.000  1.000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000  0.0000  0.6633  0.5361  1.0000  1.0000

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--- Summary for K = 9 ---

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Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.00500 0.01056 0.01278 0.01396 0.01667 0.03389

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000  0.0000  0.0400  0.1108  0.1922  0.7378
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1

```

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.00000 0.00000 0.04556 0.11079 0.20500 0.61333
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000  0.0000  0.0000  0.2095  0.2544  1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000  0.9328  1.0000  0.9268  1.0000  1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.00000 0.00000 0.00000 0.06472 0.00000 1.00000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.

```

```
0.0000 0.0000 0.1061 0.3881 0.9178 1.0000
', '\n`'))
```

```
## ‘‘‘
##
##
## --- Summary for K = 1 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.2217 0.2417 0.2483 0.2496 0.2572 0.2856
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.5761 0.7175 0.6618 0.8056 0.9761
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.5622 0.6972 0.6435 0.7839 0.9422
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.01556 0.04556 0.07111 0.23779 0.26667 1.00000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.02278 0.04556 0.06778 0.20864 0.20625 1.00000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.01722 0.05278 0.13556 0.33933 0.63833 1.00000
##
## --- Summary for K = 2 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.03944 0.05444 0.05833 0.05865 0.06278 0.08167
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
```



```

## 0.0000 0.3979 0.5628 0.5163 0.6806 0.9517
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000 0.3889 0.5483 0.4980 0.6567 0.9111
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.00000 0.01778 0.43722 0.50325 1.00000 1.00000
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.000000 0.007778 0.045556 0.329230 0.857083 1.000000
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000 0.6083 1.0000 0.7680 1.0000 1.0000
##
## --- Summary for K = 3 ---
## Z.ar:
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.004444 0.012778 0.015556 0.015614 0.018333 0.030000
##
## beta.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000 0.2817 0.4533 0.4189 0.5806 0.9178
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:

```

```

## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.0000 0.2840 0.4528 0.4107 0.5689 0.8500
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.0000 1.0000 1.0000 0.8456 1.0000 1.0000
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.0000000 0.0005556 0.0097222 0.2908248 0.7755556 1.0000000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.0000 0.5583 1.0000 0.7613 1.0000 1.0000
##
## --- Summary for K = 4 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.003889 0.011667 0.015556 0.017287 0.021667 0.050000
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.0000 0.1289 0.3292 0.3133 0.4744 0.8950
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.0000 0.1344 0.3344 0.3085 0.4700 0.8217
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.0000 0.9489 1.0000 0.8198 1.0000 1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.0000000 0.0000000 0.0005556 0.2987883 0.9131944 1.0000000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.000 1.000 1.000 0.865 1.000 1.000

```

```

##
## --- Summary for K = 5 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.001667 0.007778 0.010000 0.010944 0.013333 0.065000
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0750 0.2594 0.2602 0.4044 0.8633
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0800 0.2689 0.2584 0.4061 0.7694
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.4017 0.9417 0.6898 1.0000 1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.0000 0.2785 0.7489 1.0000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 1.0000 1.0000 0.9043 1.0000 1.0000
##
## --- Summary for K = 6 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.001667 0.005000 0.006667 0.007303 0.008889 0.029444
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0500 0.2139 0.2269 0.3572 0.8489
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

```

```

##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.05444 0.22167 0.22363 0.35778 0.72056
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.4206  0.4813  1.0000  1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.575   1.000   1.000   1.000   1.000   1.000
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.2445  0.4968  1.0000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.8367  1.0000  0.8093  1.0000  1.0000
##
## --- Summary for K = 7 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.001667 0.005556 0.007222 0.007894 0.009444 0.031667
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01389 0.19167 0.20592 0.33889 0.82556
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01556 0.20333 0.20642 0.34778 0.69167
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.2017  0.4351  1.0000  1.0000

```

```

## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.5033  1.0000  1.0000  0.9999  1.0000  1.0000
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.2216  0.2803  1.0000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.7094  1.0000  0.7798  1.0000  1.0000
##
## --- Summary for K = 8 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.001667 0.005556 0.006667 0.007356 0.008889 0.020556
##
## beta.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.002778 0.107222 0.148832 0.253333 0.771667
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.003333 0.117222 0.151073 0.268194 0.632778
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.3722  0.4600  1.0000  1.0000
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  1.0000  1.0000  0.9964  1.0000  1.0000
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000  0.000  0.000  0.139  0.000  1.000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.6633  0.5361  1.0000  1.0000
##
## --- Summary for K = 9 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00500 0.01056 0.01278 0.01396 0.01667 0.03389
##

```

```

## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.0400 0.1108 0.1922 0.7378
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.04556 0.11079 0.20500 0.61333
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.0000 0.2095 0.2544 1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.9328 1.0000 0.9268 1.0000 1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.06472 0.00000 1.00000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.1061 0.3881 0.9178 1.0000
##
## ‘ ‘ ‘

```

Conclusion

and basic statistics for each K (1–9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.